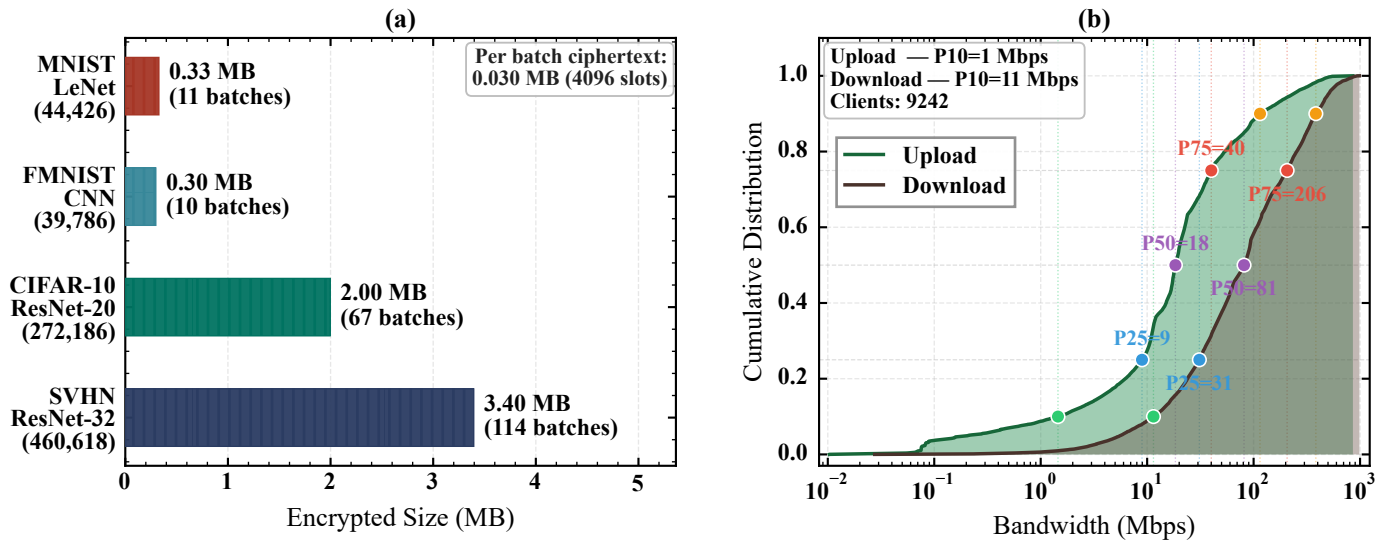

(a)
MNIST
LeNet
(44,426)
0.33 MB
(11 batches)
FMNIST
CNN
(39,786)
0.30 MB
(10 batches)
CIFAR-10
ResNet-20
(272,186)
2.00 MB
(67 batches)
SVHN
ResNet-32
(460,618)
3.40 MB
(114 batches)
Per batch ciphertext:
0.030 MB (4096 slots)
0
1
2
3
4
5
Encrypted Size (MB)
(b)
Upload — P10=1 Mbps
Download — P10=11 Mbps
Clients: 9242
Upload
Download
P75=40
P75=206
P50=18
P50=81
P25=9
P25=31
Cumulative Distribution
0.0
0.2
0.4
0.6
0.8
1.0
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Bandwidth (Mbps)